

Question number	Answer	Additional Guidance	Mark
	• Zebra fish are {not caused pain/given a painful stimulus / returned to the source to prevent {ecological change / maintain population} ;	Accept no physical damage / stress / humane conditions / not harmed Ignore no licence needed / not against the law / they are social fish / has a simple nervous system or does not feel (much) pain unqualified ignore ethical comments	(1)

Question number	Answer	Additional Guidance	Mark
i	• Correct expected value calculated (1) • Correct use of formula (1) • Correct answer (1)	No ECF 36 $\frac{44-36}{36} + \frac{28-36}{36}$ Other values can be awarded as long as two sums are being added 3.56 or 3.556 Correct answer with no working gains 3 marks 32/9 is MP 1 and 2 only	(3)

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ii	{prediction was correct / null hypothesis was accepted} as the value of chi squared is less than the critical value (1)	As we do not know their calculated value allow converse statements. Must state the calculated/chi squared value is more / less than the critical value/3.84 Accept there is no difference as the value of chi squared is less than the critical value/ore converse Ignore hypothesis unqualified	(1)

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	An answer that makes reference to one of the following: <i>abiotic</i> (water) temperature - light intensity - pH	Accept comment on water quality/oxygen concentration Accept wavelength of light	(1)

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i	- variable with suitable control method described (1) {results are not valid / description of expected effect on the dependent variable} (1)	For temperature allow <i>thermostatic</i> water bath / tank heater / air conditioned room / AC / incubator Accept thermometer with a suitable method to adjust temperature Accept an increase or decrease but not just affected/changed Accept control and effect of a biotic variable	(2)

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	so fish cannot {learn (from past experience) / become habituated}	Accept {fish could become habituated (if used more than once) (behavioural) stress / eq / might develop a preference / go to the same colour again } Ignore harm unqualified/ethical issues/explanations	(1)